AI BLAST SAFETY ASSISTANT

Enterprise Solution Presentation Deck & Executive Pitch

GFF AI ACADEMY
CAPSTONE PROJECT

[SLIDE 1] Executive Summary & Core Value Proposition

• **Project Title:** AI-Assisted Pre-Blast Safety Evaluation & Optimisation System.

• **Objective:** Modernize open-pit mine blasting operations through deterministic risk scoring and AI advisory decision-support.

• **Core Problem:** Blasting involves catastrophic hazards (flyrock, premature detonation, toxic fumes). Traditional paper checklists are prone to human error, missed safety protocols, and lack digital auditability.

• **Solution:** A hybrid 3-tier system featuring a **Deterministic Rule Engine** (calculates risk scores 0-100 & Green/Yellow/Orange/Red bands), **LLM Advisory Engine** (Google Gemini 3.1 Flash-Lite / Claude 3.5), and **Blasting Officer Digital Authorization**.

[SLIDE 2] System Architecture & Technology Stack

• **Frontend:** Single-Page Application built with React 18, Vite, TypeScript, and Tailwind CSS. Features dynamic multi-tab state preservation, canvas digital signature capture, and interactive chart visualizations.

• **Backend API:** High-performance Python FastAPI server using Pydantic request validation and Uvicorn async execution.

• **Dual Database Architecture:**

• • **SQLite (SQLAlchemy):** Role-Based Access Control (RBAC), user sessions, hashed bcrypt credentials.

• • **MongoDB (Motor Async):** High-volume JSON document storage for safety checklists and blast simulation logs.

• **AI Engine:** Auto-detecting multi-vendor LLM client (Google Gemini REST API / Anthropic Claude API) with automatic deterministic fallback.

[SLIDE 3] Module Breakdown & Key Functional Capabilities

• **Module 1 — Pre-Blast Safety Checklist:** Evaluates 17 site parameters across Weather, Shift, Workforce, Equipment, and Site conditions. Auto-populates current date and triggers mandatory RED risk status on critical hazards (lightning, workers in exclusion zone, unapproved design).

• **Module 2 — Blast Design & Fragmentation Optimisation:** Simulates peak particle velocity (PPV), air blast noise (dB), throw distance, and Rosin-Rammler rock fragmentation curves based on burden, spacing, and charge weight.

• **Module 3 — Incident Registers & Safety Auditing:** Real-time safety breach logging correlated directly with pre-blast inspection reports.

[SLIDE 4] Safety & Governance Architecture ('AI Advises, Human Decides')

• **Separation of Duties (Four-Eyes Principle):** Legally restricts official blast approval strictly to verified **Authorised Blasting Officers**. Non-authorized roles are blocked at UI and API levels (HTTP 403 Forbidden).

• **Deterministic First:** AI cannot decide or alter risk scores. Risk scoring is 100% deterministic, explainable, and reproducible.

• **Single-Page Tamper-Evident PDF Reports:** Generates a compact A4 PDF audit certificate containing complete checklist metrics, officer signature graphics, and a **SHA-256 integrity verification hash**.

[SLIDE 5] Deployment, Demo Results & Enterprise Future Scope

• **Containerization:** Fully Dockerized via multi-container `docker-compose` setup supporting local dev and production PostgreSQL/MongoDB clusters.

• **Future Expansion:** Seamless REST API integration into enterprise Mining Intelligence Platforms alongside Worker Wearable Safety Tracking and Drone Topographical Mapping.